

Full Stack

Generative AI & Agentic AI with **PYTHON**



Core Foundations

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|----------------------|----------------------|
| ➔ Python | ➔ LangChain |
| ➔ Machine Learning | ➔ CrewAI |
| ➔ Deep Learning | ➔ FastAPI |
| ➔ NLP | ➔ HuggingFace |
| ➔ LLM Fundamentals | ➔ Pinecone |
| ➔ Prompt Engineering | ➔ ChromaDB |
| ➔ Generative AI | ➔ FAISS |
| ➔ Agentic AI | ➔ Fine Tuning |
| ➔ MCP | ➔ LoRA / QLoRA |
| ➔ RAG | ➔ Cloud Deployment |
| ➔ Vector Databases | ➔ Resume Preparation |

AI IDEs

- | | |
|------------------|--------------------|
| ➔ Cursor AI | ➔ OpenAI Codex CLI |
| ➔ Windsurf | ➔ AI Coding |
| ➔ Claude Code | ➔ AI Debugging |
| ➔ GitHub Copilot | ➔ AI Refactoring |
| ➔ Gemini CLI | ➔ AI Code Review |

LangGraph & Agent Frameworks

- | | |
|-----------------------|--|
| ▪ LangGraph Basics | ▪ Memory |
| ▪ StateGraph | ▪ Supervisor, Reflection & Planning Agents |
| ▪ Nodes & Edges | ▪ OpenAI Agents SDK |
| ▪ Conditional Routing | ▪ Agno |
| ▪ Parallel Execution | ▪ AutoGen |
| ▪ Human-in-the-Loop | ▪ PydanticAI |
| ▪ Checkpointing | |

Model Providers

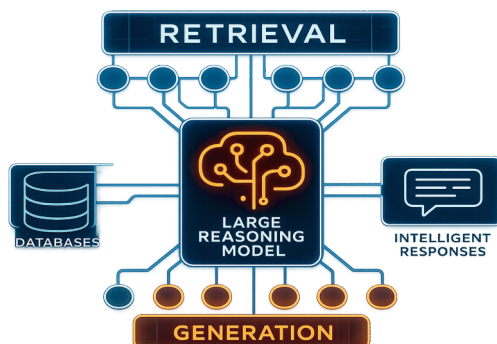
- OpenAI
- Claude
- Gemini
- DeepSeek
- Qwen
- Mistral
- Groq
- Moonshot (Kimi)
- Cohere
- xAI Grok
- Pricing
- Context Window
- Vision
- Reasoning

LLMOps

- LangSmith
- MLflow
- Arize Phoenix
- Weights & Biases
- TruLens
- Prompt Versioning
- Evaluation
- Monitoring
- Tracing
- Guardrails
- Deployment

Advanced RAG

- Hybrid Search
- Dense/Sparse Retrieval
- Rerankers
- Context Compression
- Query Expansion
- Parent-Child Retrieval
- Graph RAG
- Agentic RAG
- Corrective RAG
- Self RAG



Multimodal AI

OCR
Speech
Audio
Video
Document AI
Vision Models
GPT Vision
Gemini Vision
Claude Vision
Qwen VL

AI Security

Prompt Injection
Jailbreaks
API Security
MCP Security
Guardrails
Hallucination-
Detection
Red Teaming

Local LLMs

Ollama
LM Studio
llama.cpp

GGUF
Quantization
GPU vs CPU



Deployment & DevOps

Docker

Docker Compose

Kubernetes

Vercel

Railway

Render

Hugging Face Spaces

AWS Bedrock

Azure AI

Vertex AI

Databases & APIs

PostgreSQL

MongoDB

Redis

Neo4j

Elasticsearch

OpenAI SDK

Anthropic SDK

Gemini SDK

Groq SDK

Together AI

OpenRouter

Automation

n8n

Gmail

WhatsApp

Slack

Telegram

Google Drive

Calendar

Excel

CRM Automation



Python Advanced

Async Programming
Decorators
Generators
Context Managers
Type Hinting

Pydantic
Dataclasses
OOP
Testing
Logging

Git & Linux

Git
GitHub
Branches
PRs
GitHub Actions

CI/CD
Linux Commands
SSH
Cron
Environment Variables

FastAPI Advanced

JWT
OAuth2
Streaming

WebSockets
File Upload
Async APIs

Projects

10 Mini Projects
5 Industry Projects
1 Capstone Project
AI Resume Screener
SQL Agent

Medical Assistant
Legal Assistant
Voice Bot
Research Assistant
Marketing Agent

Interview & Placement

ATS Resume
LinkedIn
GitHub Portfolio
Kaggle
Mock Interviews
System Design
LLM Interviews

Internship

30-Day Internship
45-Day Internship
Industry Mentor
Project Reviews
Certification

Quantum Computing (Future AI)

- Quantum Computing Fundamentals
- Qubits
- Quantum Gates
- Superposition
- Entanglement
- Quantum Algorithms
- Qiskit Basics
- Quantum Machine Learning
- Quantum AI Use Cases
- Future of Quantum + Generative AI

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